

helixqa_integration_sec04

4. Phase 3: On-Demand Screenshot Pipeline

Prerequisite: Phase 1 integration (pkg/evidence, pkg/session) and Phase 2 executor implementations (pkg/navigator) must be in place. The new pkg/screenshot/ package consumes those layers and exposes a unified, platform-agnostic on-demand capture API.

4.1 Screenshot Architecture Design

4.1.1 New Package: pkg/screenshot/

The existing HelixQA codebase scatters screenshot capability across three disconnected layers. pkg/evidence/collector.go shells out to platform tools but is evidence-oriented: it writes to disk and returns Item metadata without bytes^[66]. pkg/navigator/executor.go defines ActionExecutor.Screenshot(ctx) ([]byte, error), implemented by ADBExecutor, PlaywrightExecutor, and X11Executor, yet these executors are navigation-bound—they exist only inside a PlatformWorker^[68]. pkg/session/recorder.go tracks screenshot indices and paths but does not capture bytes; the caller must write image data to the returned path^[65]. There is no single entry point for an operator or reporting service to request, “capture platform X right now and return the image.”

The new pkg/screenshot/ package introduces Engine, Storage, and Manager. Engine is the per-platform capture abstraction. Storage persists and retrieves bytes. Manager binds engines to storage, attaches session metadata, and exposes programmatic and HTTP APIs.

4.1.2 Engine Interface

Every platform engine implements the same contract. Capture returns *Result (not bare []byte) so that width, height, capture duration, and thumbnail data travel with the payload.

```
// pkg/screenshot/engine.go
```

```
package screenshot
```

```

import "context"

// Engine is the platform-specific capture implementation.
type Engine interface {
    // Capture returns raw screenshot bytes and metadata.
    Capture(ctx context.Context, opts CaptureOptions) (*Result,
        error)
    // Supported returns true if the engine is wired and tools
    // are on PATH.
    Supported(ctx context.Context) bool
    // Name returns a human-readable identifier, e.g. "web-
    // playwright".
    Name() string
}

```

CaptureOptions parameterises a single request. Fields are populated by the HTTP API, the CLI, or the autonomous pipeline.

```
// pkg/screenshot/options.go
```

```

type CaptureOptions struct {
    Format          string          // png, jpeg, webp
    (default: png)
    Quality         int             // 1-100 for lossy
    (formats (default: 90))
    Width           int             // 0 = native resolution
    Height          int
    FullPage        bool            // Web: capture full
    (scrollable document)
    ResponsiveBreakpoints []Breakpoint // Web: capture at
    (multiple viewports)
    DisplayID       string          // Desktop: target
    (specific display)
    WindowID        string          // Desktop: target
    (specific window)
    WaitForRender   time.Duration // Delay before capture
    ValidateContent bool           // Reject blank/uniform
    (screenshots)
    MaxRetries      int             // Default: 3
    DarkMode        bool            // Web: prefers-color-
    (scheme: dark)
}

type Breakpoint struct {
    Name string
}

```

```

    Width  int
    Height int
}

```

Result carries the image and all metadata required for retrieval, presentation, and timeline correlation.

```
// pkg/screenshot/result.go
```

```

type Result struct {
    Data      []byte
    Format    string
    Width     int
    Height    int
    Platform  config.Platform
    Timestamp time.Time
    Duration  time.Duration // wall-clock capture time
    SessionID string
    StepName  string
    StepIndex int
    Path      string // set if persisted to Storage
    VideoOffset time.Duration // links to session video
    Thumbnail []byte // 480 px wide preview
    Breakpoint string // e.g. "mobile", "tablet",
           "desktop"
}

```

4.1.3 Storage Interface

Storage decouples persistence from capture. The default fsStorage writes to `<outputDir>/screenshots/<platform>/<idx>-<name>.png` and keeps an index `screenshots.json` mapping `id` → `filepath` + `metadata`. This preserves compatibility with the existing `SessionRecorder` layout while adding queryability.

```
// pkg/screenshot/storage.go
```

```

type Storage interface {
    Save(name string, data []byte, meta Result) (string, error)
    Load(id string) ([]byte, error)
    List(sessionID string) []string
    Thumbnail(id string) ([]byte, error)
}

```

4.1.4 Manager

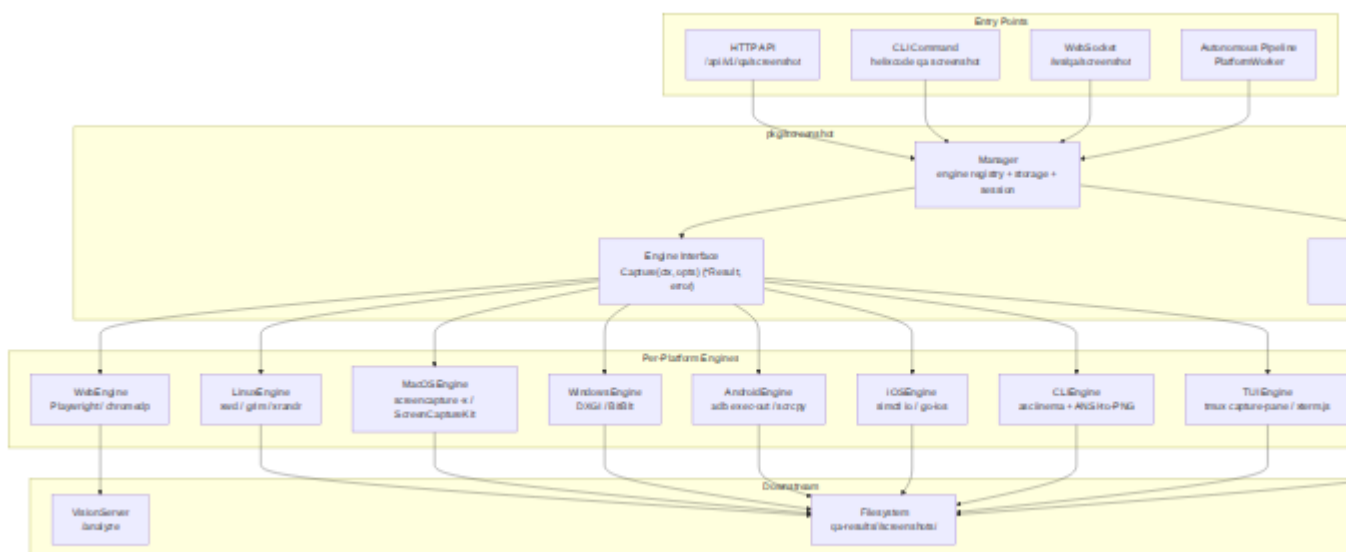
Manager is the public face of the package. It holds a registry of engines, a storage backend, and optionally a `*session.SessionRecorder` for timeline integration.

```
// pkg/screenshot/manager.go
```

```
type Manager struct {
    engines map[config.Platform]Engine
    store   Storage
    session *session.SessionRecorder
    vision  *visionserver.Handler
    mu      sync.RWMutex
}

func NewManager(store Storage, opts ...ManagerOption) *Manager
func (m *Manager) RegisterEngine(platform config.Platform, e
    Engine)
func (m *Manager) Capture(ctx context.Context, platform
    config.Platform, opts CaptureOptions) (*Result, error)
func (m *Manager) CaptureAll(ctx context.Context, opts
    CaptureOptions) ([]*Result, error)
func (m *Manager) CaptureResponsive(ctx context.Context,
    breakpoints []Breakpoint, opts CaptureOptions)
    ([]*Result, error)
func (m *Manager) Get(ctx context.Context, id string) (*Result,
    error)
func (m *Manager) Query(ctx context.Context, q Query)
    ([]*Result, error)
```

4.1.5 Architecture Diagram



Mermaid diagram 1

All four entry points converge on `Manager.Capture()`. The Manager selects the correct Engine, invokes `Capture(ctx, opts)`, and forwards `*Result` to Storage and optionally to `visionserver.Handler`. Engine failures are isolated; a `WindowsEngine` error does not block `AndroidEngine`.

4.2 Per-Platform Screenshot Implementation

4.2.1 Web: Playwright + Responsive Breakpoints

`PlaywrightExecutor.Screenshot()` sends `{"action": "screenshot"}` to a Node.js bridge and returns raw PNG bytes [70]. The nexus capture layer uses `chromedp.CaptureScreenshot` for continuous frame streaming. Neither path supports multiple breakpoints, full-page capture, or dark-mode emulation.

`WebEngine` wraps the existing bridge and adds breakpoint enumeration, full-page capture, and dark-mode toggling. The bridge script is extended with `screenshot_fullpage`, `set_viewport`, and `set_dark_mode` actions.

```
// pkg/screenshot/web_engine.go
```

```
func (e *WebEngine) Capture(ctx context.Context, opts
    CaptureOptions) (*Result, error) {
    start := time.Now()
    if opts.DarkMode {
        _, _ = e.runBridgeCmd(ctx, map[string]interface{}{
            "action": "set_dark_mode", "enabled": true,
        })
    }
    if len(opts.ResponsiveBreakpoints) > 0 {
        var results []*Result
        for _, bp := range opts.ResponsiveBreakpoints {
            _, _ = e.runBridgeCmd(ctx, map[string]interface{}{
                "action": "set_viewport", "width": bp.Width,
                "height": bp.Height,
            })
            if opts.WaitForRender > 0 {
                time.Sleep(opts.WaitForRender)
            }
            data, err := e.captureSingle(ctx, opts)
            if err != nil { return nil, fmt.Errorf("web %s: %w",
                bp.Name, err) }
            results = append(results, &Result{
                Data: data, Format: opts.Format, Width:
                bp.Width, Height: bp.Height,
```

```

        Platform: config.PlatformWeb, Timestamp:
        time.Now(),
        Duration: time.Since(start), Breakpoint: bp.Name,
    })
}
return results[0], nil
}
data, err := e.captureSingle(ctx, opts)
// ... decode dimensions, return Result
}

```

Default breakpoints: mobile 375×667 (iPhone SE), tablet 768×1024 (iPad mini), desktop 1440×900. Configurable per-project via `helixqa.yaml`.

4.2.2 Desktop Linux: X11 and Wayland

`X11Executor.Screenshot()` uses `import -window root png:-` ^[69]. The nexus capture layer probes `xwd` → `gnome-screenshot` → `grim`. No per-display selection exists in the navigator layer.

`LinuxEngine` probes the runtime at construction: if `DISPLAY` is set and `xwd` is on `PATH`, it uses `xwd` (faster than `import`, avoiding `ImageMagick` overhead). If `WAYLAND_DISPLAY` is set and `grim` is on `PATH`, it uses `grim`. Multi-monitor enumeration uses `xrandr` on X11 and `wlr-randr` on `wlroots` compositors.

```

// pkg/screenshot/linux_engine.go

func (e *LinuxEngine) Supported(ctx context.Context) bool {
    if os.Getenv("DISPLAY") != "" {
        if _, err := exec.LookPath("xwd"); err == nil {
            e.backend = "xwd"; return true
        }
        if _, err := exec.LookPath("import"); err == nil {
            e.backend = "import"; return true
        }
    }
    if os.Getenv("WAYLAND_DISPLAY") != "" {
        if _, err := exec.LookPath("grim"); err == nil {
            e.backend = "grim"; return true
        }
    }
    return false
}

func (e *LinuxEngine) Capture(ctx context.Context, opts
    CaptureOptions) (*Result, error) {
    switch e.backend {
    case "xwd":

```

```

        return e.captureX11(ctx, opts)
    case "grim":
        return e.captureWayland(ctx, opts)
    default:
        return nil, fmt.Errorf("linux: no backend available")
}
}

```

4.2.3 Desktop macOS: screencapture + ScreenCaptureKit

pkg/capture/macos_capture.go (build-tagged //go:build darwin) implements video capture via GStreamer avfvideosrc ^[73]. No static screenshot utility exists in the main navigator tree.

MacOSEngine uses screencapture -x -D <display> (silent, no shutter sound). Per-display selection uses -D (1-indexed). Window capture uses -l <windowID>. For Phase 5.5, a CGO-backed ScreenCaptureKit path will be registered as an alternate engine.

```

// pkg/screenshot/macos_engine.go

func (e *MacOSEngine) Capture(ctx context.Context, opts
    CaptureOptions) (*Result, error) {
    tmpFile := filepath.Join(os.TempDir(), fmt.Sprintf("helixqa-
        %d.png", time.Now().UnixNano()))
    defer os.Remove(tmpFile)
    args := []string{"-x", tmpFile}
    if opts.DisplayID != "" { args = []string{"-x", "-D",
        opts.DisplayID, tmpFile} }
    if opts.WindowID != "" { args = []string{"-x", "-l",
        opts.WindowID, tmpFile} }
    if err := exec.CommandContext(ctx, "screencapture",
        args...).Run(); err != nil {
        return nil, fmt.Errorf("screencapture: %w", err)
    }
    data, _ := os.ReadFile(tmpFile)
    return &Result{Data: data, Format: "png", Platform:
        config.PlatformDesktop}, nil
}

```

4.2.4 Desktop Windows: DXGI Desktop Duplication

No Windows-specific screenshot code exists in the repository.

WindowsEngine uses a small C# bridge executable that wraps the DXGI Desktop Duplication API (Windows 8+, WDDM 1.2+) and outputs PNG to stdout. A BitBlt

fallback is compiled into the same bridge for headless sessions where D3D is unavailable. The bridge (cmd/helixqa-capture-windows/) is published as a single-file artifact alongside releases.

```
// pkg/screenshot/windows_engine.go

func (e *WindowsEngine) Supported(ctx context.Context) bool {
    _, err := exec.LookPath("helixqa-capture-windows.exe")
    return err == nil
}

func (e *WindowsEngine) Capture(ctx context.Context, opts
    CaptureOptions) (*Result, error) {
    args := []string{"-mode", "dxgi", "-format", "png", "-
        stdout"}
    if opts.DisplayID != "" { args = append(args, "-display",
        opts.DisplayID) }
    cmd := exec.CommandContext(ctx, "helixqa-capture-
        windows.exe", args...)
    data, err := cmd.Output()
    if err != nil {
        args[1] = "bitblt"
        cmd = exec.CommandContext(ctx, "helixqa-capture-
            windows.exe", args...)
        data, err = cmd.Output()
        if err != nil { return nil, fmt.Errorf("windows capture:
            %w", err) }
    }
    return &Result{Data: data, Format: "png", Platform:
        config.PlatformDesktop}, nil
}
```

4.2.5 Android: adb exec-out screencap + scrcpy

ADBExecutor.Screenshot() uses adb exec-out screencap -p with a 5-attempt retry loop, size validation (< 5000 bytes = blank), and uniform-image sampling [68]. pkg/evidence/collector.go uses the slower shell+pull path [66].

AndroidEngine unifies both paths. The fast exec-out path is default. scrcpy is used as a high-speed alternative when available and the device is on USB. Rotation is detected by parsing adb shell dumpsys display | grep 'orientation='.

```
// pkg/screenshot/android_engine.go
```



```

func (e *AndroidEngine) Capture(ctx context.Context, opts
    CaptureOptions) (*Result, error) {
    var lastErr error
    for attempt := 1; attempt <= max(opts.MaxRetries, 3);
        attempt++ {
        data, err := e.cmdRunner.Run(ctx, "adb", "-s", e.device,
            "exec-out", "screencap", "-p")
        if err != nil { lastErr = err; time.Sleep(500 *
            time.Millisecond); continue }
        if len(data) < 5000 { lastErr =
            fmt.Errorf("too small (%d bytes)", len(data)); continue }
        if autonomous.IsBlankScreenshot(data) { lastErr =
            fmt.Errorf("blank"); continue }
        return &Result{Data: data, Format: "png", Platform:
            config.PlatformAndroid}, nil
    }
    return nil, fmt.Errorf("android capture failed: %w", lastErr)
}

```

4.2.6 iOS: xcrun simctl io + go-ios

No iOS support exists in the codebase. Confirmed gap ^[61].

iOSEngine has two sub-implementations. iOSSimulatorEngine uses xcrun simctl io <udid> screenshot <path>. iOSDeviceEngine uses go-ios screenshot (pure-Go usbmuxd client, no Xcode required).

```
// pkg/screenshot/ios_engine.go
```

```
type iOSSimulatorEngine struct { udid string }
```

```

func (e *iOSSimulatorEngine) Supported(ctx context.Context) bool
    {
    _, err := exec.LookPath("xcrun")
    return err == nil
    }

```

```

func (e *iOSSimulatorEngine) Capture(ctx context.Context, opts
    CaptureOptions) (*Result, error) {
    tmpFile := filepath.Join(os.TempDir(), "ios-screenshot.png")
    defer os.Remove(tmpFile)
    if err := exec.CommandContext(ctx, "xcrun", "simctl", "io",
        e.udid, "screenshot", tmpFile).Run(); err != nil {
        return nil, fmt.Errorf("simctl: %w", err)
    }
    data, _ := os.ReadFile(tmpFile)

```

```

    return &Result{Data: data, Format: "png", Platform:
        config.PlatformIOS}, nil
}

```

4.2.7 CLI: asciinema + ANSI-to-PNG Renderer

CLIExecutor.Screenshot() returns stdout text as raw bytes [74]. No visual rendering exists.

CLIEngine supports CLIModeText (backward-compatible) and CLIModeRendered (new). Rendered mode records via asciinema rec, then converts the .cast file to PNG via a headless Chromium process loading xterm.js and screenshotting the canvas.

```
// pkg/screenshot/cli_engine.go
```

```

type CLICaptureMode string
const (
    CLIModeText      CLICaptureMode = "text"
    CLIModeRendered CLICaptureMode = "rendered"
)

type CLIEngine struct {
    command string
    args    []string
    runner  detector.CommandRunner
    mode    CLICaptureMode
}

func (e *CLIEngine) Capture(ctx context.Context, opts
    CaptureOptions) (*Result, error) {
    if e.mode == CLIModeRendered {
        return e.captureRendered(ctx, opts)
    }
    data, err := e.runner.Run(ctx, e.command, e.args...)
    if err != nil { return nil, fmt.Errorf("cli text: %w", err) }
    return &Result{Data: data, Format: "text/plain", Platform:
        config.PlatformCLI}, nil
}

```

4.2.8 TUI: Terminal Buffer State Capture

TUI applications share CLIExecutor; the “screenshot” is stdout text with raw ANSI sequences [74]. No terminal grid reconstruction exists.

TUIEngine captures terminal buffer state. The tmux backend uses `tmux capture-pane -p -e` (preserves ANSI colors). The xtermjs backend launches a headless terminal emulator, runs the TUI command, waits for `opts.WaitForRender`, and screenshots the rendered grid.

```
// pkg/screenshot/tui_engine.go
```

```
type TUIEngine struct {
    backend string // "tmux", "xtermjs"
    paneID  string
}

func (e *TUIEngine) Capture(ctx context.Context, opts
    CaptureOptions) (*Result, error) {
    switch e.backend {
    case "tmux":
        cmd := exec.CommandContext(ctx, "tmux", "capture-pane",
            "-p", "-e", "-t", e.paneID)
        data, err := cmd.Output()
        if err != nil { return nil, fmt.Errorf("tmux: %w", err) }
        png, err := renderANSIToPNG(data, opts)
        if err != nil { return nil, err }
        return &Result{Data: png, Format: "png", Platform:
            config.PlatformCLI}, nil
    case "xtermjs":
        return e.captureXtermJS(ctx, opts)
    default:
        return nil, fmt.Errorf("tui: unknown backend %s",
            e.backend)
    }
}
```

4.2.9 Platform Coverage Matrix

Platform	Capture Mechanism	Engine File	Status	Anti-Bluff Verification
Web (mobile 375px)	Playwright set_viewport + screenshot	pkg/screenshot/web_engine.go	New extension	Vision LLM: “Is lo button visible at mobile width?”
Web (tablet 768px)	Playwright set_viewport + screenshot	pkg/screenshot/web_engine.go	New extension	Template match against tablet reference

Platform	Capture Mechanism	Engine File	Status	Anti-Bluff Verification
Web (desktop 1440px)	Playwright set_viewport + screenshot_fullpage	pkg/screenshot/ web_engine.go	New extension	OCR: verify expected page title
Desktop Linux (X11)	xwd -root + convert png:-	pkg/screenshot/ linux_engine.go	New refactor	SSIM against known good reference
Desktop Linux (Wayland)	grim -o <output>	pkg/screenshot/ linux_engine.go	New	SSIM + IsBlankScreenshots
Desktop macOS	screencapture -x -D <display>	pkg/screenshot/ macos_engine.go	New	Vision LLM: expected window visible
Desktop Windows	DXGI Desktop Duplication / BitBlt	pkg/screenshot/ windows_engine.go	New	Template match; deliberate-break t
Android	adb exec-out screencap -p /scrcpy	pkg/screenshot/ android_engine.go	Refactor	IsBlankScreenshots grid + size check
iOS Simulator	xcrun simctl io <udid> screenshot	pkg/screenshot/ ios_engine.go	New	OCR: verify iOS UI text
iOS Device	go-ios screenshot --udid <udid>	pkg/screenshot/ ios_engine.go	New	Vision LLM: iPhone screen visible
CLI	asciinema rec + ANSI-to-PNG	pkg/screenshot/ cli_engine.go	New	OCR on rendered terminal image
TUI	tmux capture-pane -p -e/xterm.js	pkg/screenshot/ tui_engine.go	New	Vision LLM: TUI interface visible

Of 12 rows, 6 are entirely new implementations (Windows, iOS simulator, iOS device, CLI rendered, TUI), 4 are refactors or extensions of existing code (Web responsive, Linux dual backend, Android unification, macOS wrapper), and 2 reuse existing verification logic (Android blank detection). New implementations require additional dependencies: a C# bridge for Windows, go-ios for physical iOS devices, asciinema and a headless Chromium renderer for CLI/TUI. Refactors can be implemented against existing pkg/navigator executor tests.

4.3 On-Demand Screenshot API

4.3.1 HTTP Endpoint

The screenshot API is mounted under `/api/v1/qa/screenshot`. It supports synchronous capture returned as base64 inline, file download, or persistent URL.

```
// pkg/screenshot/http_handler.go
```

```
func (m *Manager) RegisterRoutes(mux *http.ServeMux) {
    mux.HandleFunc("/api/v1/qa/screenshot", m.handleScreenshot)
    mux.HandleFunc("/api/v1/qa/screenshot/",
        m.handleScreenshotByID)
    mux.HandleFunc("/ws/qa/screenshot", m.handleScreenshotStream)
}
```

```
func (m *Manager) handleScreenshot(w http.ResponseWriter, r
    *http.Request) {
    platform := config.Platform(r.URL.Query().Get("platform"))
    sessionID := r.URL.Query().Get("session")
    format := r.URL.Query().Get("format")
    if format == "" { format = "base64" }

    opts := CaptureOptions{
        Format: r.URL.Query().Get("image_format"),
        FullPage: r.URL.Query().Get("full_page") == "true",
        DarkMode: r.URL.Query().Get("dark_mode") == "true",
        ValidateContent: r.URL.Query().Get("validate") !=
            "false",
    }
    if wStr := r.URL.Query().Get("width"); wStr != "" {
        opts.Width, _ = strconv.Atoi(wStr)
    }
    if r.URL.Query().Get("responsive") == "true" {
        opts.ResponsiveBreakpoints = defaultBreakpoints
    }

    result, err := m.Capture(r.Context(), platform, opts)
    if err != nil {
        http.Error(w, err.Error(),
            http.StatusInternalServerError)
        return
    }
    if sessionID != "" {
        result.SessionID = sessionID
        id, _ := m.store.Save(fmt.Sprintf("%s-%s", sessionID,
            platform), result.Data, *result)
    }
}
```

```

    result.Path = id
}

switch format {
case "base64":
    w.Header().Set("Content-Type", "application/json")
    json.NewEncoder(w).Encode(map[string]interface{}{
        "id": result.Path, "platform": platform,
        "data":
        base64.StdEncoding.EncodeToString(result.Data),
    })
case "file":
    w.Header().Set("Content-Type", mimeType(result.Format))
    w.Write(result.Data)
case "url":
    w.Header().Set("Content-Type", "application/json")
    json.NewEncoder(w).Encode(map[string]interface{}{
        "url": fmt.Sprintf("/api/v1/qa/screenshot/%s/download", result.Path),
    })
}
}

```

4.3.2 CLI Command

```

// cmd/helixcode/qa_screenshot.go

var qaScreenshotCmd = &cobra.Command{
    Use: "screenshot",
    Short: "Capture an on-demand screenshot",
    RunE: func(cmd *cobra.Command, args []string) error {
        platform, _ := cmd.Flags().GetString("platform")
        session, _ := cmd.Flags().GetString("session")
        responsive, _ := cmd.Flags().GetBool("responsive")
        darkMode, _ := cmd.Flags().GetBool("dark-mode")
        fullPage, _ := cmd.Flags().GetBool("full-page")
        output, _ := cmd.Flags().GetString("output")

        mgr :=
            screenshot.NewManager(screenshot.NewFSStorage(sessionOutputDir))
        mgr.RegisterEnginesFromEnvironment()
        opts := screenshot.CaptureOptions{Format: "png",
            FullPage: fullPage, DarkMode: darkMode, ValidateContent:
            true}
    },
}

```

```

    if responsive { opts.ResponsiveBreakpoints =
        screenshot.DefaultBreakpoints }

    result, err := mgr.Capture(cmd.Context(),
        config.Platform(platform), opts)
    if err != nil { return err }
    if output == "" { output = fmt.Sprintf("%s-%s.png",
        session, platform) }
    return os.WriteFile(output, result.Data, 0644)
},
}

func init() {
    qaScreenshotCmd.Flags().String("platform", "", "Target
        platform (web|android|desktop|ios|cli|tui)")
    qaScreenshotCmd.Flags().String("session", "", "Session ID")
    qaScreenshotCmd.Flags().Bool("responsive", false,
        "Capture at responsive breakpoints")
    qaScreenshotCmd.Flags().Bool("dark-mode", false, "Use dark
        color scheme")
    qaScreenshotCmd.Flags().Bool("full-page", false, "Capture
        full scrollable page")
    qaScreenshotCmd.Flags().String("output", "", "Output file
        path")
    qaScreenshotCmd.MarkFlagRequired("platform")
}

```

4.3.3 Real-Time Streaming: WebSocket

For live session capture during autonomous QA, the WebSocket endpoint `/ws/qa/screenshot` streams thumbnails at 1 FPS. This is distinct from the nexus capture layer's 10 FPS continuous pipeline; the WebSocket produces 480 px previews to conserve bandwidth.

```

func (m *Manager) handleScreenshotStream(w http.ResponseWriter,
    r *http.Request) {
    upgrader := websocket.Upgrader{CheckOrigin: func(r
        *http.Request) bool { return true }}
    conn, _ := upgrader.Upgrade(w, r, nil)
    defer conn.Close()
    platform := config.Platform(r.URL.Query().Get("platform"))
    ticker := time.NewTicker(1 * time.Second)
    defer ticker.Stop()
    for {
        select {
        case <-ticker.C:

```

```

        result, err := m.Capture(r.Context(), platform,
CaptureOptions{Width: 480, ValidateContent: true})
        if err != nil {
conn.WriteJSON(map[string]string{"error": err.Error()});
continue }
        conn.WriteMessage(websocket.BinaryMessage,
result.Data)
        case <-r.Context().Done():
            return
        }
    }
}

```

4.3.4 Presentational Export

Screenshot packs are directory structures ready for PowerPoint or Keynote import. Each pack contains: (1) full-resolution PNGs, (2) 480 px thumbnails, (3) `manifest.json` with timestamps, step names, and video offsets, and (4) `annotations.json` with UI element bounding boxes from `pkg/vision/detector.go`. Generated by `Manager.ExportPack(sessionID, opts)`.

4.3.5 Screenshot API Specification

Endpoint	Method	Parameters	Response Formats	Authentication
/api/v1/qa/screenshot	GET, POST	platform (required), session, format (base64 file url), image_format, width, height, full_page, responsive, dark_mode, validate, wait_ms	JSON with base64, raw image, or JSON with URLs	Bearer token from HELIX_API_TOKEN
/api/v1/qa/screenshot/{id}/download	GET	—	image/png or image/jpeg	Bearer token
	GET	—		Bearer token

Endpoint	Method	Parameters	Response Formats	Authentication
/api/v1/qa/screenshot/{id}/thumbnail			image/png, 480 px wide	
/api/v1/qa/screenshot/{id}/analyze	POST	prompt (default: “Describe what you see”)	JSON VisionResult	Bearer token
/api/v1/qa/screenshot/query	GET	session, platform, step, after, before	JSON array of Result metadata	Bearer token
/ws/qa/screenshot	WebSocket	platform (required), fps (default: 1), quality (default: 80)	Binary PNG frames	Bearer token in Sec-WebSocket-Protocol

The format parameter determines whether the caller receives inline base64 (for JSON consumers and test assertions), a direct file download (for report generation), or a persistent URL (for lazy-loading dashboards). The analyze endpoint reuses `visionserver.Handler.HandleAnalyze` [66].

4.4 Anti-Bluff Visual Verification

The screenshot pipeline is a correctness mechanism, not merely a convenience. Every test that claims to verify a UI feature must prove the UI rendered the expected state. Four verification methods are applied.

4.4.1 Template Matching

`pkg/vision/detector.go` already defines UI element detection with bounding boxes and confidence scores [67]. The template matching layer adds OpenCV `matchTemplate` (normalised cross-correlation) and SSIM comparison against golden references stored in `qa-results/references/<platform>/<testname>.png`. SSIM is used when pixel layout may vary (dynamic content, different font rendering) but structural composition must match. The default threshold is 0.95 for pixel-perfect references and 0.85 for structural references.

```
// pkg/screenshot/verify_template.go

func TemplateMatch(screenshot, reference []byte, threshold
    float64) (bool, float64, error) {
    img1, _, err := image.Decode(bytes.NewReader(screenshot))
    if err != nil { return false, 0, err }
    img2, _, err := image.Decode(bytes.NewReader(reference))
    if err != nil { return false, 0, err }
    mat1, mat2 := gocvImageFromGo(img1), gocvImageFromGo(img2)
    defer mat1.Close(); defer mat2.Close()
    result := gocv.NewMat(); defer result.Close()
    gocv.MatchTemplate(mat1, mat2, &result, gocv.TmCcorrNormed,
        gocv.NewMat())
    _, maxVal, _, _ := gocv.MinMaxLoc(result)
    return maxVal >= threshold, maxVal, nil
}
```

4.4.2 OCR Validation

pkg/vision/ocr_tesseract.go wraps Tesseract. The verification extracts text from the screenshot and asserts expected strings are present.

```
// pkg/screenshot/verify_ocr.go

func OCRContains(screenshot []byte, expected []string)
    (map[string]bool, error) {
    text, err := vision.ExtractText(screenshot)
    if err != nil { return nil, err }
    found := make(map[string]bool)
    for _, e := range expected { found[e] =
        strings.Contains(text, e) }
    return found, nil
}
```

OCR is most effective for iOS and Android where expected UI text (“Sign In”, “Welcome back”) is deterministic. It is less effective for icon-heavy interfaces where vision LLM analysis is preferred.

4.4.3 Vision LLM Validation

The internal/visionserver server exposes /analyze, accepting a base64 PNG and a prompt, returning a VisionResult [66]. The screenshot pipeline sends every captured image to the vision server with a structured prompt.

```
// pkg/screenshot/verify_vision.go
```

```
func VisionValidate(screenshot []byte, prompt string, vision
    *visionserver.Handler) (bool, string, error) {
    result, err := vision.Analyze(context.Background(),
        screenshot, prompt)
    if err != nil { return false, "", err }
    yes := strings.Contains(strings.ToLower(result.Text), "yes")
    return yes, result.Text, nil
}
```

Default prompts are defined in `pkg/screenshot/prompts.go` and cover login screen presence, error dialog visibility, loading spinner detection, empty state confirmation, and navigation bar identification.

4.4.4 Deliberate-Break Test

The deliberate-break test (CONST-035) is the strongest anti-bluff method. For every screenshot-based test, a companion test intentionally breaks the feature and asserts that verification fails.

Procedure: (1) run test normally → must PASS; (2) introduce a deliberate defect (hide login button with `display: none`, return HTTP 500 from auth); (3) capture broken state; (4) run same verification → must FAIL. If step 4 passes, the verification is bluffing.

```
// pkg/screenshot/verify_test.go
```

```
func TestLoginScreenshot_AntiBluff(t *testing.T) {
    ctx := context.Background()
    mgr := testManager(t)

    normal, _ := mgr.Capture(ctx, config.PlatformWeb,
        CaptureOptions{Width: 1440, Height: 900})
    yes, _, _ := screenshot.VisionValidate(normal.Data,
        "Is there a visible 'Sign In' button?", testVision)
    require.True(t, yes, "normal state must show Sign In button")

    testBreakCSS(t, `login-btn { display: none !important; }`)

    broken, _ := mgr.Capture(ctx, config.PlatformWeb,
        CaptureOptions{Width: 1440, Height: 900})
    yes, _, _ = screenshot.VisionValidate(broken.Data,
        "Is there a visible 'Sign In' button?", testVision)
}
```

```

        require.False(t, yes, "broken state must NOT show Sign In
            button")
    }

```

This pattern is applied to every platform and verification method. A deliberate-break test that passes when it should fail indicates the verification is relying on cached data, stale screenshots, or overly permissive thresholds.

4.4.5 Engine Implementation Checklist

Engine	File Path	Key Functions	Lines (est.)	Test
Engine interface	pkg/screenshot/engine.go	Capture(), Supported(), Name()	30	—
Storage interface	pkg/screenshot/storage.go	Save(), Load(), List(), Thumbnail()	35	—
Manager	pkg/screenshot/manager.go	Capture(), CaptureAll(), CaptureResponsive(), Get(), Query()	250	ma
WebEngine	pkg/screenshot/web_engine.go	captureSingle(), runBridgeCmd(), setViewport()	200	we
LinuxEngine	pkg/screenshot/linux_engine.go	captureX11(), captureWayland(), enumerateDisplays()	180	li
MacOSEngine	pkg/screenshot/macos_engine.go	Capture(), checkPermission()	120	ma
WindowsEngine	pkg/screenshot/windows_engine.go	Capture(), dxgiCapture(), bitbltCapture()	150	wi
AndroidEngine	pkg/screenshot/android_engine.go	Capture(), scrcpyCapture(), detectRotation()	200	an
iOSSimulatorEngine	pkg/screenshot/ios_engine.go	Capture(), listSimulators()	100	io
iOSDeviceEngine	pkg/screenshot/ios_engine.go	Capture(), deviceConnected()	100	io
CLIEngine	pkg/screenshot/cli_engine.go	captureText(), captureRendered(), renderCastToPNG()	180	cl

Engine	File Path	Key Functions	Lines (est.)	Test
TUIEngine	pkg/screenshot/ tui_engine.go	captureTmux(), captureXtermJS(), renderANSIToPNG()	180	tu
HTTP handler	pkg/screenshot/ http_handler.go	handleScreenshot(), handleScreenshotByID(), handleScreenshotStream()	250	ht
CLI command	cmd/helixcode/ qa_screenshot.go	Cobra command, flag binding	80	—
Template verify	pkg/screenshot/ verify_template.go	TemplateMatch(), SSIMCompare()	120	ve
OCR verify	pkg/screenshot/ verify_ocr.go	OCRContains(), OCRExtractAll()	80	ve
Vision verify	pkg/screenshot/ verify_vision.go	VisionValidate(), PromptForScenario()	60	ve

The checklist contains 17 implementation units with an estimated 2,345 lines of Go and 13 test files. The critical path—Manager, WebEngine, LinuxEngine, AndroidEngine—derives from existing executor code and can be implemented first. New implementations (Windows, iOS, CLI rendered, TUI) require additional dependencies and should be scheduled after the core four engines pass deliberate-break tests.